

FillMyPDF v2

High-Level Architecture & Feature Reference

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FillMyPDF is a production-ready, AI-powered PDF auto-fill SaaS platform. It lets users store encrypted data profiles and automatically fill PDF forms in bulk using AI vision mapping — with support for sync/async jobs, template libraries, e-signatures, webhooks, Chrome extension, and Zapier integration.

- HR Onboarding
- Tax Forms
- Real Estate
- Pharma PA Forms

50+ API Endpoints	30+ Built-in Templates	4 Worker Threads	3 Cache Layers
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Tech Stack

Layer	Technology
Backend Framework	FastAPI 0.104+ / Uvicorn / Python 3.12
Data Validation	Pydantic v2.5+
PDF Processing	pypdf, pdf2image, Pillow, reportlab, commonforms
AI / Vision	Gemini 2.5 Flash (OpenAI-compatible endpoint)
Security	AES-256-GCM encryption, bcrypt, slowapi
Storage	File-based JSON (no relational database)
Async Jobs	ThreadPoolExecutor (4 workers)
Integrations	Chrome Extension (MV3), Zapier Platform CLI
Deployment	Docker (Python 3.12-slim) + Fly.io (3 GB volume)
Testing	pytest + pytest-asyncio + pytest-cov (28 files)

System Architecture

CLIENT LAYER

REST API Clients • Chrome Extension (MV3) • Zapier Integration



API GATEWAY LAYER

RequestID Middleware → CORS → Rate Limiter (slowapi, per-tier)
Auth: X-API-Key → bcrypt validation → tier lookup



ROUTE LAYER (FastAPI)

/profiles /batch /templates /jobs /extract
/signing /keys /pdf /billing /health



SERVICE LAYER

ProfileService BatchFillService TemplateService APIKeyService
JobRunner VisionService PDFService ExtractionService
SigningSessionService EmailService BillingService



REPOSITORY LAYER

ProfileRepository APIKeyRepository TemplateRepository JobRepository



PERSISTENT STORAGE (File-based JSON)

profiles/ (AES-256-GCM) • api_keys/ (bcrypt) • jobs/ • templates/ • usage_stats.json



External Services

Gemini 2.5 Flash (OpenAI-compatible API) • Stripe (billing, optional) • SMTP (e-sign emails, optional) • Outbound Webhooks (HMAC-SHA256 signed, 4 retries)

Core PDF Fill Pipeline

1 INPUT NORMALIZATION

User data (JSON / CSV / XLSX) is normalized by InputAdapter into the Canonical Model — Person, Provider, Organization, or Medication domains — reducing field-mapping ambiguity.

2 PDF FILLABILITY CHECK

PDFService checks whether the uploaded PDF already has AcroForm fields.

- YES → use directly
- NO → commonforms ML model (FFDNet-S) converts static PDF to fillable; result cached as fillable.pdf for all future fills

3 LABEL DETECTION (pdfplumber)

VisionService reads field bounding boxes (pypdf) then uses pdfplumber to find nearby words on the same horizontal band:

- Textbox → look LEFT (≤ 220 px, 4 words)
- Checkbox → look RIGHT (≤ 500 px, 6 words)

Falls back to raw field name if no text found.

4 THREE-LAYER CACHE CHECK

Layer 1: In-memory per-request cache

Layer 2: Disk cache (TemplateCache JSON — indexed by template+field)

Layer 3: AI call to Gemini 2.5 Flash ← only on cache miss

Cache hit → deterministic fills, zero AI cost, zero variance

5 AI FIELD MAPPING (Gemini 2.5 Flash)

Labeled field list + PDF page image → Gemini 2.5 Flash.

Returns: { field_name → value, confidence_score }

Fields below FILL_CONFIDENCE_THRESHOLD are dropped (not filled).

Result written back to disk cache for future requests.

6 WRITE & OUTPUT

PDFService writes values into AcroForm fields (pypdf / reportlab).

Output: ZIP archive (batch) or single PDF stream → client.

Async Job Architecture

POST /api/v1/jobs/batch



Returns 202 Accepted immediately



JobRepository: persist job (status: queued)



ThreadPoolExecutor: 4 concurrent workers



Run full batch fill pipeline per record



Update progress: {total, completed, successful, failed}



On complete: POST webhook (HMAC-SHA256, 4 retries exp. backoff)



Client polls **GET /api/v1/jobs/{job_id}** for progress, or listens for the HMAC-signed webhook POST. Manual replay available via **POST /api/v1/jobs/{job_id}/webhook-redelivery**.

Security Architecture

API Keys

Format: fmp_live_<160-bit base32>

Stored: bcrypt hash (cost 12 prod, 4 test)

Shown: plaintext once at creation only

Tiers: free / pro / business / admin

Revoke: DELETE /api/v1/keys/{id}

Profile Data

Algorithm: AES-256-GCM

Key deriv: PBKDF2-SHA256 (100k iters)

Per-field: random salt + nonce

Auto-detect sensitive fields:

SSN, EIN, DOB, card numbers

Names/emails stored in plaintext

Webhooks & API

Webhook sig: HMAC-SHA256

Header: X-FillMyPDF-Signature

Stripe IPN: verified server-side

Rate limits: per-tier via slowapi

CORS: explicit origins (no wildcard)

Request IDs: full trace correlation

Subscription Tiers & Rate Limits

Tier	Profiles	Requests / min	Requests / day
free	1	60	10,000
pro	unlimited	600	100,000
business	unlimited	6,000	1,000,000
admin	unlimited	100,000	unlimited

Accuracy Features

❖ Coordinate-based Label Detection

Uses pdfplumber to find actual on-page label text next to each field before calling AI. Avoids relying on internal field names like *field_0001*.

❖ Direction-Aware Scanning

Textboxes look LEFT (labeled on left side); far-left checkboxes look RIGHT — mirrors real PDF layout conventions for highest label accuracy.

❖ AI Confidence Threshold

FILL_CONFIDENCE_THRESHOLD (default 0.0, recommended 0.5) — fields mapped below the score are skipped rather than filled with wrong data. Prevents noisy or hallucinated fills.

❖ Gemini 2.5 Flash (Multimodal)

Handles visual + text context simultaneously. Receives labeled field list AND a rendered page image for richer spatial understanding.

❖ Deterministic Template Cache

Once a form is mapped, results are cached to disk. Subsequent fills are fully deterministic — no AI variance, zero extra cost, faster turnaround.

❖ Configurable DPI

PDF-to-image rendered at 200 DPI by default. Higher DPI settings produce sharper images for improved AI vision quality on complex forms.

❖ Canonical Data Model

Input normalized to Person / Provider / Org / Med domains before mapping — reduces field-matching ambiguity by providing structured semantic context.

❖ AcroForm-Native Path

If the PDF already has AcroForm fields, fills them natively. AI is only used for field→data mapping, not for reading the PDF — no OCR guesswork.

Accuracy Configuration Knobs (.env)

```
FILL_CONFIDENCE_THRESHOLD=0.5 # skip low-confidence fields (0.0 = allow all)

DEFAULT_DPI=300 # higher = sharper images for AI vision

COMMONFORMS_CONFIDENCE=0.1 # static-to-fillable detection sensitivity

DEFAULT_AI_MODEL=gemini-2.5-flash # swap to more capable model if needed

TEMPLATE_CACHE_ENABLED=true # persist mapping cache to disk
```

TEMPLATE_CACHE_TTL_DAYS=0 # 0 = cache never expires

Complete Feature Matrix

Core Processing

Sync batch fill — JSON / CSV / XLSX	✓
Static PDF → AcroForm conversion (ML)	✓
AcroForm field extraction to JSON/CSV	✓
AI field mapping (Gemini 2.5 Flash)	✓
Confidence threshold filtering	✓
PDF merge & split utilities	✓

Async & Jobs

Async job queue (202/poll pattern)	✓
4 concurrent ThreadPool workers	✓
Job progress tracking	✓
Outbound webhooks (HMAC-SHA256)	✓
Webhook auto-retry (exp. backoff × 4)	✓
Manual webhook redelivery	✓

Templates & Caching

Pre-built template library (30+ forms)	✓
Template upload/manage (admin)	✓
3-layer mapping cache (memory+disk)	✓
Lazy fillable PDF conversion + cache	✓
Template batch fill	✓

Security & Auth

API key auth (X-API-Key)	✓
bcrypt key hashing (cost 12)	✓
AES-256-GCM profile encryption	✓
PBKDF2 key derivation (100k iters)	✓
Per-tier rate limiting (slowapi)	✓
Request ID correlation	✓

Integrations

Chrome Extension (Manifest V3)	✓
Zapier Platform CLI integration	✓
Stripe billing (checkout + portal)	✓

SMTP e-sign notifications	✓
OpenAPI docs (Swagger + ReDoc)	✓
E-Signatures	
Visual signature overlay (PNG stamp)	✓
Multi-signature session management	✓
Signature audit log	✓
Certificate-based PAdES signing	—

API Quick Reference

System (no auth)

GET	/	Service info
GET	/health	Health check
GET	/usage	Global usage stats

API Keys (admin only)

POST	/api/v1/keys	Create API key
GET	/api/v1/keys	List keys
DELETE	/api/v1/keys/{id}	Revoke key

Profiles

POST	/api/v1/profiles	Create profile
GET	/api/v1/profiles	List profiles
GET	/api/v1/profiles/{id}	Get profile
PATCH	/api/v1/profiles/{id}	Update profile
DELETE	/api/v1/profiles/{id}	Delete profile
POST	/api/v1/profiles/import	Bulk CSV/Excel import

Batch (sync)

POST	/api/v1/batch/fill-json	Fill from JSON array
POST	/api/v1/batch/fill-csv	Fill from CSV
POST	/api/v1/batch/fill-xlsx	Fill from Excel
GET	/api/v1/batch/download/{file}	Download result ZIP

Async Jobs

POST	/api/v1/jobs/batch	Submit async batch (202)
GET	/api/v1/jobs/{id}	Poll job status
POST	/api/v1/jobs/{id}/webhook-redelivery	Replay webhook
DELETE	/api/v1/jobs/{id}	Cancel job

Templates

GET	/api/v1/templates	List templates
POST	/api/v1/templates/{id}/fill	Single-record fill
POST	/api/v1/templates/{id}/fill-batch	Batch fill
POST	/api/v1/templates/upload	Upload template (admin)

Extract / Sign / PDF

POST	/api/v1/extract/fields	AcroForm → JSON
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POST	/api/v1/extract/csv	AcroForm → CSV
POST	/api/v1/signing/overlay	Visual signature stamp
POST	/api/v1/pdf/merge	Merge PDFs
POST	/api/v1/pdf/split	Split PDF

Key Environment Variables

```
# Security (REQUIRED in production)

PROFILES_ENCRYPTION_KEY=<32+ char random key>

# Server

API_HOST=0.0.0.0

API_PORT=8000

DEBUG=False

# AI / Vision

DEFAULT_AI_BASE_URL=https://generativelanguage.googleapis.com/v1beta/openai/

DEFAULT_AI_MODEL=gemini-2.5-flash

DEFAULT_DPI=200

# Accuracy

FILL_CONFIDENCE_THRESHOLD=0.5 # skip low-confidence AI mappings

# Async Jobs

JOB_WORKER_THREADS=4

# Webhooks

WEBHOOK_SIGNING_SECRET=<optional HMAC secret>

WEBHOOK_MAX_ATTEMPTS=4

WEBHOOK_RETRY_BASE_DELAY_SEC=1.0

# Stripe (optional)

STRIPE_SECRET_KEY=sk_...

STRIPE_PUBLISHABLE_KEY=pk_...

STRIPE_WEBHOOK_SECRET=whsec_...

# SMTP – e-sign notifications (optional)

SMTP_HOST=smtp.example.com

SMTP_PORT=587
```

SMTP_USERNAME=user@example.com

SMTP_PASSWORD=secret

APP_BASE_URL=https://fillmypdf.example.com